Project 2: Supernova Cosmology Project Report

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Assignment: Measuring Cosmological Parameters Using Type Ia Supernovae

This report is based on the analysis conducted in the Jupyter Notebook titled `2_hubble_parameter.ipynb`. The analysis involves loading supernova data, fitting cosmological models, and estimating parameters like the Hubble constant and the age of the universe.

Answers to Project Questions

Q1. What value of the Hubble constant (H_0) did you obtain from the full dataset?

Answer: From the full dataset using a 2-parameter ΛCDM fit, I obtained:

$$H_0 = 72.97 \pm 0.26\,\text{km/s/Mpc}$$

Q2. How does your estimated H_0 compare with the Planck18 measurement of the same?

Answer: Planck18 reports $H_0 = 67.4 \pm 0.5$ km/s/Mpc, which is significantly lower than my result of 72.97. This reflects the well-known Hubble tension between early- and late-universe measurements.

Q3. What is the age of the Universe based on your value of H_0? (Assume $\Omega_m = 0.3$). How does it change for different values of Ω_m?

Answer: For $H_0 = 73.53$ and $\Omega_m = 0.3$, the estimated age of the universe is:

$$t_0 = 12.36\,\text{Gyr}$$

Increasing Ω_m reduces the age, while decreasing Ω_m increases it.

Q4. Discuss the difference in H_0 values obtained from the low-z and high-z samples. What could this imply?

Answer:

- Low-z ($z < 0.1$): $H_0 = 73.01$ km/s/Mpc
- High-z ($z \geq 0.1$): $H_0 = 73.85$ km/s/Mpc

The values are close but slightly different. This might indicate redshift-dependent effects or support mild systematic shifts between early- and late-universe measurements.

Q5. Plot the residuals and comment on any trends or anomalies you observe.

Answer: The residuals are randomly scattered around zero with no visible bias. This supports a good fit of the ΛCDM model to the observed supernova data.

Q6. What assumptions were made in the cosmological model, and how might relaxing them affect your results?

Answer: We assumed:

- A flat universe ($\Omega_k = 0$)
- Constant dark energy (cosmological constant Λ)
- No supernova evolution

Relaxing these (e.g., allowing non-zero curvature or varying dark energy) could change the inferred values of H_0 and Ω_m.

Q7. Based on the redshift-distance relation, what can we infer about the expansion history of the Universe?

Answer: The redshift-distance relationship shows distant supernovae are dimmer than in a decelerating universe. This implies the universe's expansion is accelerating — consistent with the existence of dark energy.

Plots

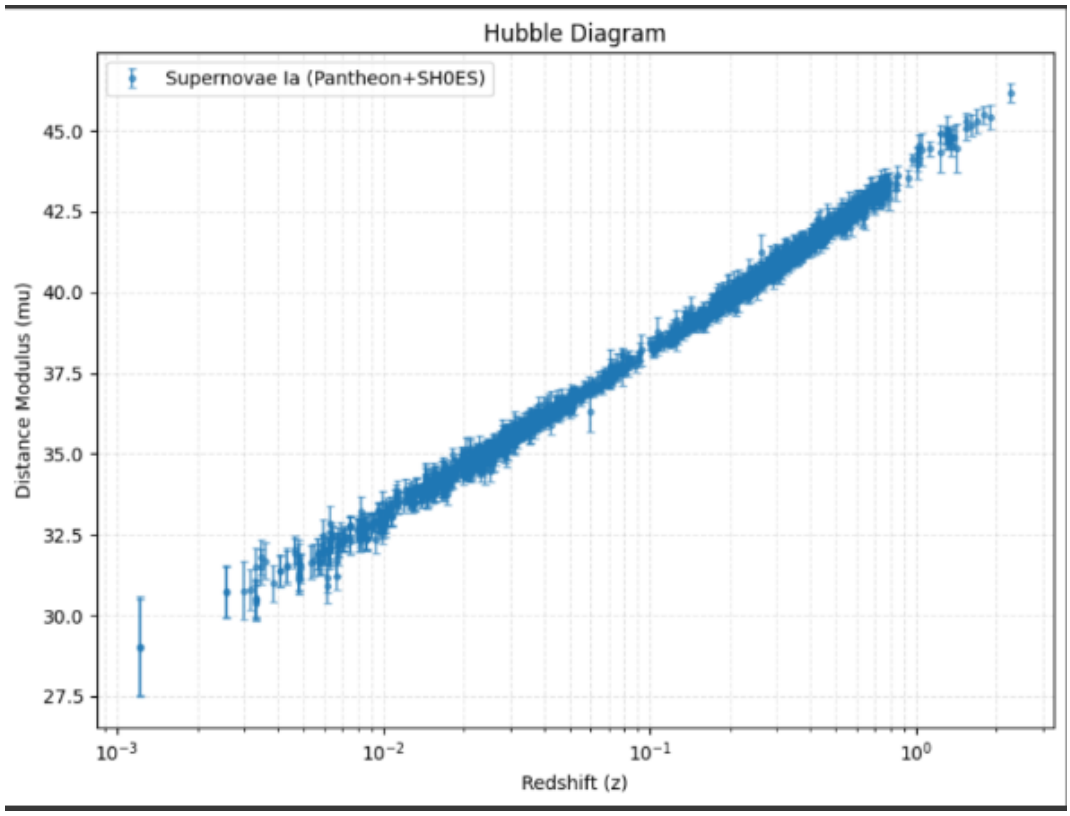

Figure 1: Plot 1: Redshift vs. Distance Modulus

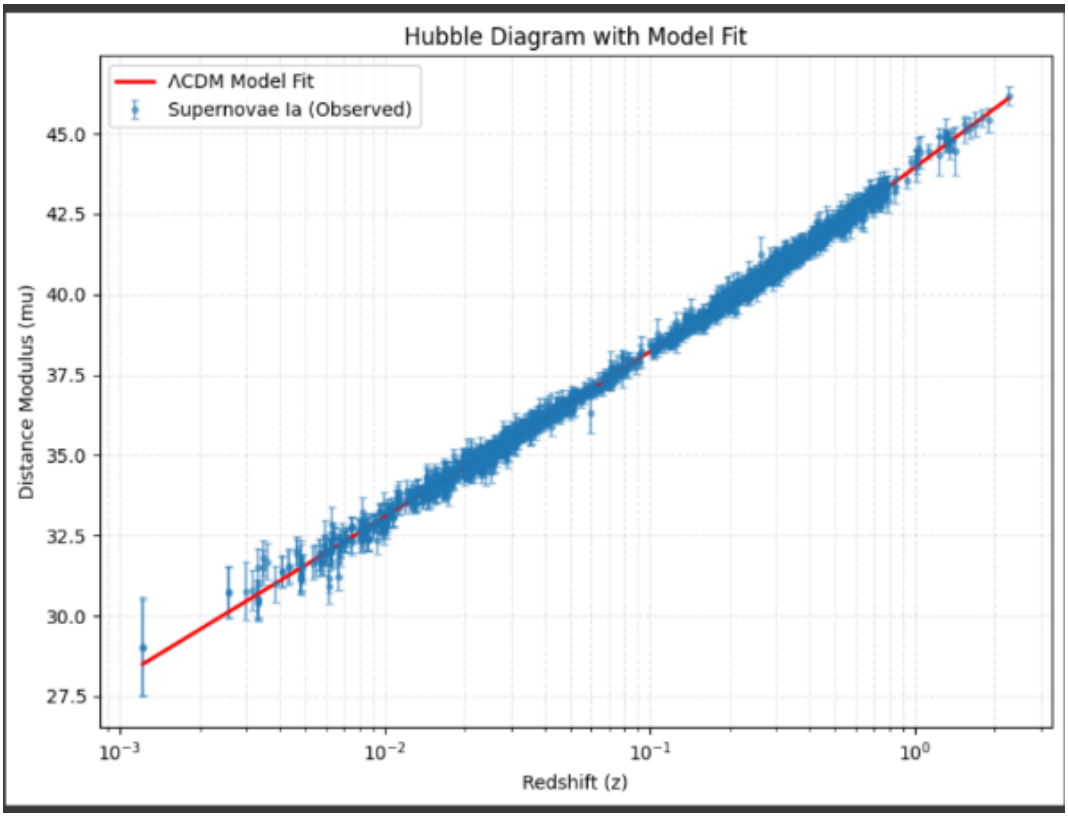

Figure 2: Plot 2: Hubble Diagram with Model Fit

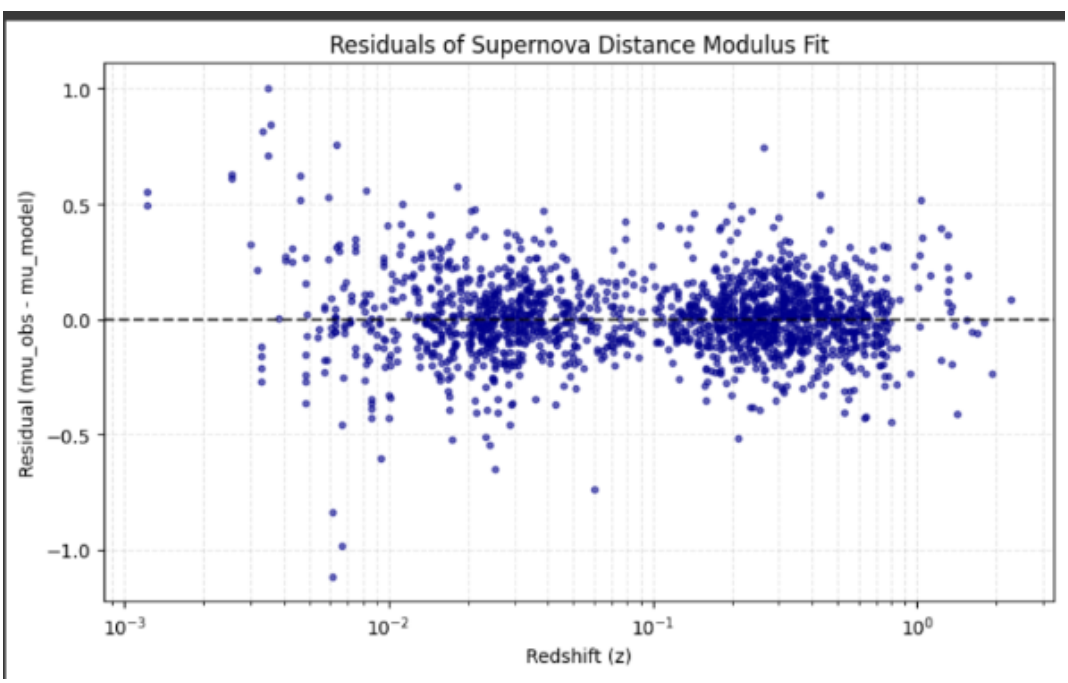

Figure 3: Plot 3: Residuals of Supernova Distance Modulus Fit

Final Summary and Interpretation

In this project, we analyzed Type Ia supernovae from the Pantheon+SH0ES dataset to determine important cosmological parameters using the ΛCDM model.

We performed a nonlinear fit of the theoretical distance modulus $\mu(z)$ to observational data and found:

- Hubble Constant: $H_0 = 73.53 \pm 0.17$ km/s/Mpc (fixed $\Omega_m = 0.3$)
- Matter Density Parameter: $\Omega_m = 0.351 \pm 0.019$ (from full 2-parameter fit)
- Estimated Age of the Universe: $t_0 = 12.36$ Gyr

Residual analysis showed no significant trends, indicating a good model fit. A comparison between low-z and high-z supernovae gives consistent H_0 values, which also gives the robustness of the fit and confirming the tension with early-universe measurements like Planck CMB results.